1. Project Introduction

E-Commerce Product Management REST API Testing Using Fake Store API, Postman, and Newman CLI

Overview:

In today's digital era, e-commerce applications rely heavily on REST APIs for communication between frontend applications, mobile applications, and backend servers. APIs are responsible for managing important business operations such as product management, user management, category management, cart operations, authentication, and order processing. Therefore, ensuring the quality, reliability, and performance of APIs is a critical part of software testing.

This project focuses on testing the Fake Store REST API using Postman and Newman CLI. Fake Store API is a publicly available RESTful web service that simulates a real-world e-commerce platform. It provides endpoints for products, categories, carts, users, and authentication, allowing testers and developers to perform various API operations without requiring access to a production environment.

The main purpose of this project is to validate the functionality of different API endpoints by sending HTTP requests and verifying the responses returned by the server. The testing process includes validating status codes, response bodies, response times, JSON structures, and environment variables. Both positive and negative test scenarios can be executed to ensure that the APIs behave as expected under different conditions.

Postman is used as the primary API testing tool for creating requests, organizing collections, managing environments, and writing automated test scripts. JavaScript assertions are added to validate API responses automatically. The Collection Runner feature of Postman is used to execute multiple requests and verify the complete workflow of the application.

To automate collection execution and generate reports, Newman CLI is integrated into the project. Newman is the command-line collection runner for Postman that allows API tests to be executed outside the Postman application. It helps generate execution reports, identify failed test cases, and analyze API performance.

The project covers the following major modules:

Authentication Module

- Login User
- Token Generation Validation

Products Module

- Get All Products
- Get Single Product
- Add New Product
- Update Product using PUT Method

- Update Product using PATCH Method
- Delete Product

Categories Module

- Get All Categories
- Get Products by Category

Cart Module

- Get All Carts
- Get Single Cart
- Add Product to Cart
- Update Cart using PUT Method
- Update Cart using PATCH Method
- Delete Cart

Users Module

- Get All Users
- Get Single User

For every API request, validations are performed to ensure that the server returns the expected response. The project includes verification of response status codes, response messages, response data accuracy, response times, and JSON schema structures. Environment variables are used to manage reusable values such as Base URL, Product ID, User ID, and Cart ID.

The project also demonstrates real-world API testing practices such as collection creation, environment management, automated validations, collection execution, Newman integration, report generation, defect identification, and result analysis.

By completing this project, testers can gain practical experience in REST API testing, Postman scripting, automated API validation, Newman execution, and report generation. These skills are widely used in software testing and quality assurance projects across the IT industry.
